Supplementary material

From routine cine CMR to regional myocardial dysfunction

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Abbreviations. AHA, American Heart Association; bSSFP, balanced steady-state free precession; CAD, coronary artery disease; CoV, coefficient of variation; DCM and HCM, dilated and hypertrophic cardiomyopathy; DENSE, displacement encoding with stimulated echoes; DL, deep learning; ED and ES, end-diastole and end-systole; Ecc, Err and Ell, circumferential, radial and longitudinal strain; EF, ejection fraction; EPE, end-point error; FT, feature tracking; ICC, intraclass correlation coefficient; LGE, late gadolinium enhancement; LoA, limits of agreement; LS, CS and RS, longitudinal, circumferential and radial strain; LV and RV, left and right ventricle; MI, myocardial infarction; NV, not verified at the available reading depth; SAX and LAX, short- and long-axis; SNR, signal-to-noise ratio; TOST, two one-sided tests; WMA, wall-motion abnormality.

Supplementary Methods: literature search, screening and evidence handling

Review type and records. This is a critical narrative review. No registered protocol, duplicate screening, meta-analysis or PRISMA flow is claimed. The search record describes a structured search, citation expansion and subsequent targeted updates. The preserved query log and extraction notes establish part of that history; missing original exports and record-level decisions prevent full reconstruction of the reported screening process.

Core set and citation expansion. The search record identifies a core set including DeepStrain and its evaluations [1–3], StrainNet [4], multivendor and material-sensitive comparisons [5–8], consensus documents [9–12] and reviews [13–17]. It describes following references and citing articles, manually adding guideline and resource records, and stopping when no method family, comparison axis or main conclusion changed. The round-by-round expansion record is unavailable, so the stopping rule is retained as a historical description; it has not been reproduced as a saturation assessment.

Historical structured search. The surviving query log records PubMed, Europe PMC and OpenAlex queries on 3 September 2026 with a publication cutoff of 1 September 2026. Six concept families were queried in each database: regional cine strain after infarction; known-motion or digital-phantom validation; prior or regularisation perturbation; conditional recoverability; failure-aware or selective prediction; and regional localisation, spillover or attenuation. The 18 exact queries, hit counts and dates are preserved in `supplementary/historical/gap-search-query-log.csv`. Recorded hit totals are 41, 244 and 508, respectively. Retrieval was capped at 100 records per query, truncating the broad post-infarction query in Europe PMC (239 hits) and OpenAlex (473 hits). The per-query returned counts sum to 281. The historical

notes report 253 unique records after deduplication, but the original exports and screening file needed to recompute that count are not present.

Historical screening and retention. The notes describe one software-assisted title-and-abstract screening pass and no full-text exclusion stage. The intended design combined prescribed regional abnormality, known motion and controlled estimator changes; records addressing part of that design were also retained. The reported counts are 21 retained and 232 set aside, followed by two additions on 9 September under the same publication cutoff (Righetti and colleagues, 2025; Yang and colleagues, 2025, preprint). A preserved 23-row near-neighbour matrix includes two versions of MulViMotion and was described as 22 studies. These are inherited counts and classifications; no new screening was performed. Individual decisions, deduplication and the historical conclusion that no retained study combined every design element cannot be re-audited from the available files. That joint criterion would not in any event establish absence of studies addressing a simpler subset. The revised main text therefore treats graded focal recovery as a proposed priority, without using this record to claim field-wide absence.

October updates and present verification. The search notes date a targeted verification pass to 8 October 2026, whereas the accompanying bibliographic ledger is named `evidence/crossref_verification_20261009.csv`. Sources published after 1 September, including the September Jahromi study and October IG-BiReg record, belong to this later update, whose complete query and inclusion history is unavailable. On 11 October, central numerical and methodological claims were checked against accessible primary-source sections and official resource pages. That pass recorded its targeted queries, direct URL retrievals, access failures and reading depth; it was not a rerun of the historical search. The existing independent identity audit covers 94 references; identity retrieval does not certify every detailed claim or a full-text reading of every source.

Reading depth and unresolved detail. The inherited 135-row claim ledger labels 75 entries full text, 56 abstract and 4 official record. It covers 69 citation keys, which do not match the complete current 94-reference set; many locators refer to source files that are no longer available. Those labels are retained as historical provenance; they do not represent current verification. In assigning reading depth, this revision distinguished inspected full-text sections, abstracts, metadata, official records, inherited extraction and failed access. Supplementary Table S3 preserves method-specific summaries at those available depths. NV denotes detail not verified by this review; it is not a claim that the source omitted it. A bibliographic-only entry is not assigned a validation result.

Software assistance. Search support, evidence organisation, drafting, revision, consistency checks, figure rendering and compilation were assisted by OpenAI Codex and Claude Code (Anthropic), as declared in the main text. AI-assisted review of intermediate drafts is not independent human screening or clinical review. Verification of all content, and responsibility for it, remain with the author.

Supplementary Note 1: strain definitions and comparison traps

Let $\boldsymbol{\varphi} : \Omega_0 \to \Omega_t$ map a reference position $\boldsymbol{X}$ to its deformed position $\boldsymbol{x}$. The displacement is

$$\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{\varphi}(\boldsymbol{X}, t) - \boldsymbol{X}, \tag{1}$$

and the deformation gradient is

$$\boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}} = \boldsymbol{I} + \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}}. \tag{2}$$

Common finite-strain measures include the right Cauchy–Green tensor $\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}$, Green–Lagrange strain $\boldsymbol{E} = (\boldsymbol{C} - \boldsymbol{I})/2$, and logarithmic strain obtained from a stretch tensor. In one dimension, engineering strain $e = (\ell - \ell_0)/\ell_0$ and Green–Lagrange strain satisfy $E = e + e^2/2$. Numerically comparing values without converting the definitions can create an apparent method bias [7].

Cardiac components require a coordinate frame. Circumferential, radial and longitudinal axes can be defined at the reference state, updated through time or constructed from a ventricular centreline. Local axes become unstable near the apex and can vary with segmentation. A transmural average differs from endocardial strain because deformation varies through the wall. Two-dimensional short-axis analyses omit longitudinal and through-plane derivatives; a reported radial or circumferential component is consequently conditional on the 2D model.

Peak definitions also differ. If $E_s(t)$ is the segment-average strain curve, $\min_t E_s(t)$ is generally not equal to the spatial average of pointwise minima. "End-systolic", "peak systolic" and "peak over the whole cardiac cycle" may diverge in dyssynchrony or post-systolic shortening. A reproducible comparison must fix the order of differentiation, projection, spatial averaging and temporal peak selection. Supplementary Figure S1 illustrates these four traps.

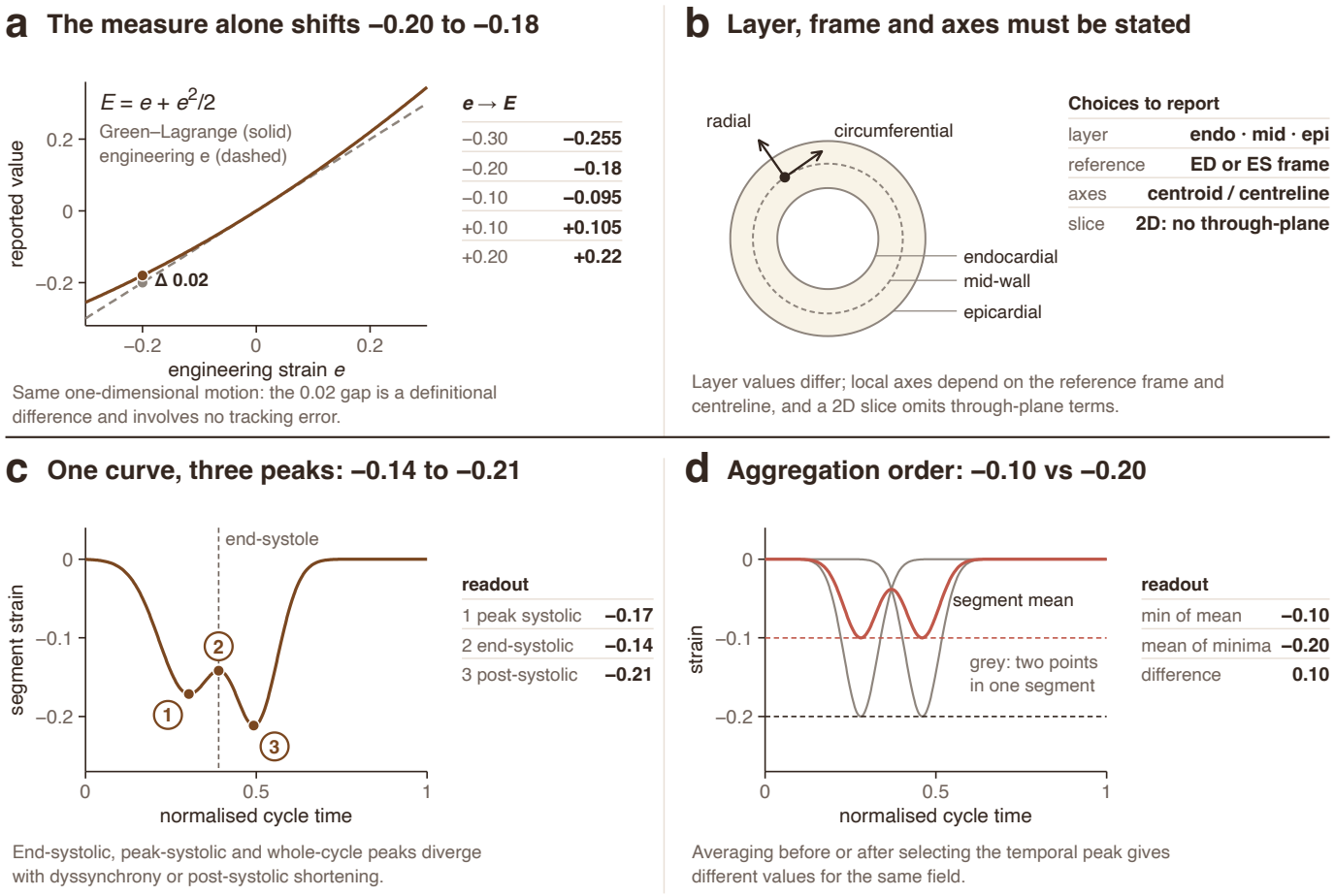


Figure S1. The same motion gives different numbers under different definitions. (a) Engineering strain e and Green–Lagrange strain $E = e + e^2/2$ differ by 0.02 at $e = -0.20$; the table converts five values. (b) Endocardial, mid-wall and epicardial layers and local radial and circumferential axes, with the choices a report must state. (c) One segmental curve read as peak systolic (−0.17), end-systolic (−0.14) and post-systolic peak (−0.21). (d) The minimum of a segment-mean curve (−0.10) differs from the mean of pointwise minima (−0.20). Conceptual and analytic schematics with no measured scale; printed values are computed from the drawn curves.

Supplementary Note 2: validation targets

The claim boundaries for common validation targets are summarised in Figure 5a of the main text.

A validation hierarchy should not be interpreted as a single ranking. Simulation supplies exact truth but imperfect realism. DENSE and tagging provide material-sensitive in-vivo information but have sequence- and processing-specific error. Scan–rescan establishes precision. LGE and outcomes test biological or clinical relationships. A mature claim often needs several targets because their weaknesses are complementary.

When two acquisitions are compared, acquisition registration is part of the reference. Slice position, breath-hold, loading and heart-rate differences can create apparent algorithm error. Where reference processing includes smoothing or temporal fitting, its effective resolution should be reported. Segment-level analyses require participant-aware uncertainty estimates because segments from one heart are correlated.

Supplementary Note 3: error attributes and mapping validity

The distinct spatial, temporal and field-validity failure modes are illustrated in Supplementary Figure S2.

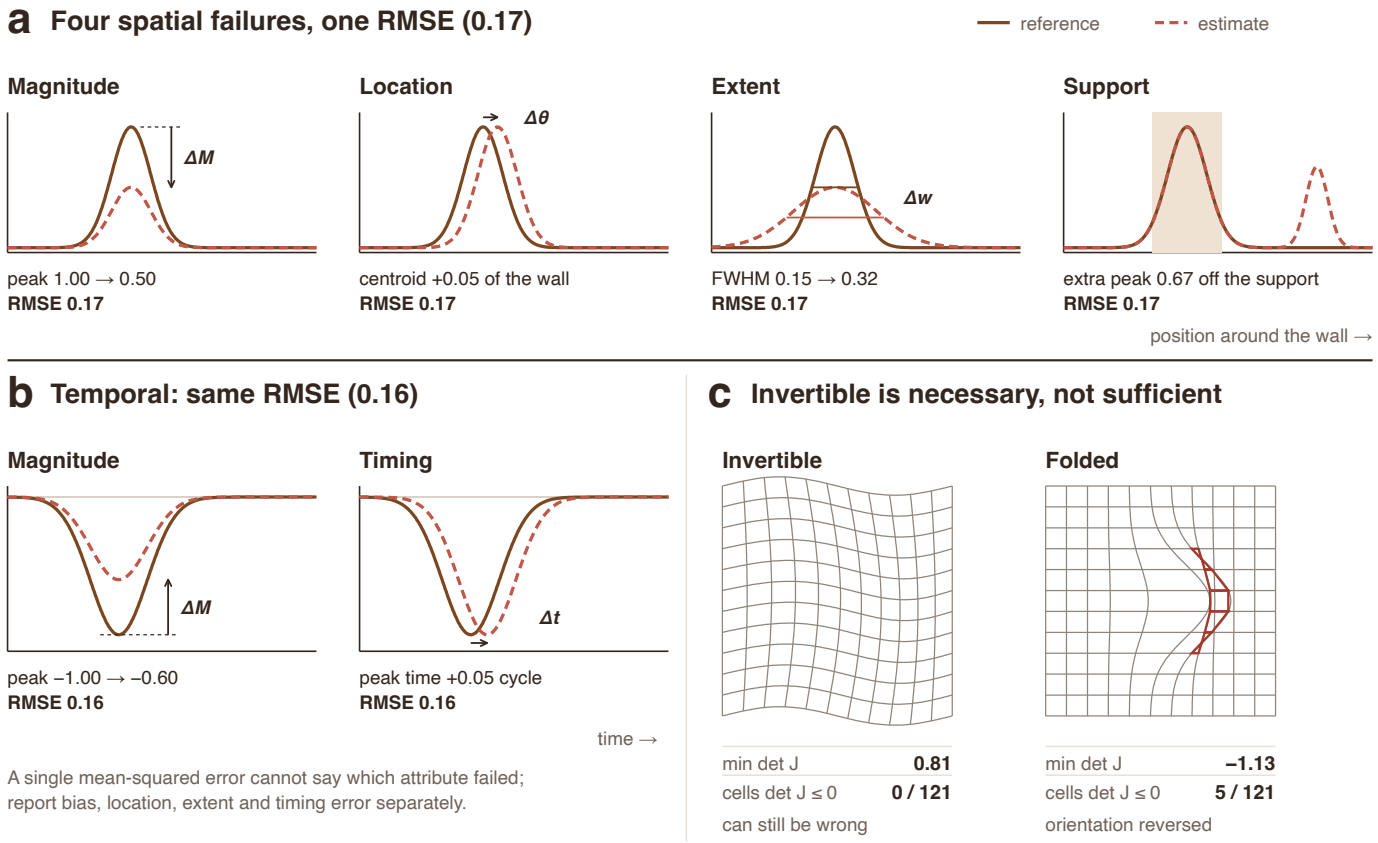


Figure S2. Regional error attributes and mapping validity. (a) Reference and estimate profiles separating magnitude, location, extent and support errors. (b) Temporal magnitude and timing errors. In a, and again in b, every estimate has the same root-mean-square error against its reference (0.17 and 0.16), so a single mean-squared error cannot tell these failures apart. (c) A smooth invertible mapping (minimum $\det J = 0.81$), which can still be wrong, and a folded mapping with $\det J \leq 0$ in 5 of 121 outlined cells (minimum -1.13). Prescribed schematics with no measured scale.

Let a prescribed abnormality field be $g_{\mathrm{ref}}(\boldsymbol{x}, t)$ and an estimate be $g_{\mathrm{est}}(\boldsymbol{x}, t)$. A single mean-squared error can conceal clinically different failures. Attribute-specific losses may include:

- magnitude error in an agreed region;
- distance between abnormality centroids or peaks;

- width, area or overlap error of a thresholded abnormal region;
- onset, peak-time or curve-alignment error; and
- error conditional on the reference support and a sensitivity analysis over the threshold.

Field validity is related but distinct. A positive Jacobian determinant is a useful check against local folding, while near-incompressibility may be physiologically motivated. Neither proves accurate correspondence. A smooth, invertible mapping can be wrong, and a reference generated by an approximate 2D model can violate three-dimensional tissue behaviour. Validity metrics should accompany regional accuracy metrics and cannot replace them.

Supplementary Note 4: learning target and control point

DeepStrain and StrainNet illustrate why the target matters. DeepStrain demonstrated a complete automated workflow with segmentation, motion estimation, volumetric agreement, reference ranges and repeatability analyses [1]. Its cine motion network was not independently validated against dense material displacement for every regional claim. StrainNet used DENSE phase-derived displacement as a supervised target, but it reported two distinct evaluations [4]. In the 62-person internal participant-held-out test set, its 0.75 ± 0.35-mm pixelwise EPE was obtained with DENSE-magnitude contours supplied to the network and compared with DENSE displacement over myocardium and time. Separately, bSSFP cine feature-tracking contours were used for the end-systolic global and AHA-segmental circumferential-strain comparison in 59 test participants. Thus the EPE and cine-contour ICCs must not be fused into one "cine displacement" result; population, contour source, registration, direction, aggregation and time point still bound the claim.

Training-time and inference-time controls must also be distinguished. Changing a loss weight usually requires matched retraining, and varying it only at test time has no effect unless the trained model exposes that parameter. Post-processing strength or a biomechanical solve, by contrast, can be a genuine inference-time control. Evaluation should compare like with like, hold splits and estimands fixed, and keep final test cases out of model or rule selection.

Supplementary Figure S3 shows the two control points and their shared held-out evaluation contract.

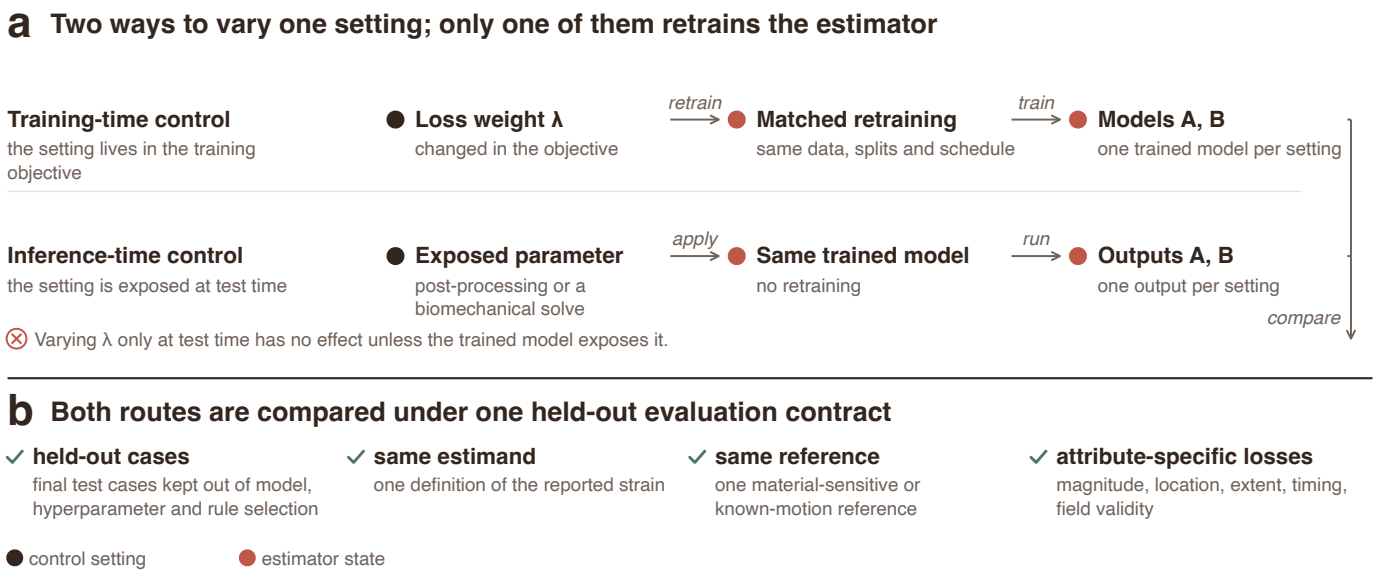


Figure S3. Training-time versus inference-time controls. A training-loss weight requires matched retraining; a test-time parameter can be varied only when the implementation exposes it. Candidate models or outputs are compared on the same held-out cases under the same estimand, reference and attribute-specific losses. Conceptual schematic.

Supplementary Note 5: pre-specified evaluation template

For each intended-use claim, the following items can be completed before the final test set is analysed:

Item	Pre-specified content
Intended use	Clinical/research decision; user; consequence of false regional call
Population	Disease, age, rhythm, prior treatment, exclusions and recruitment setting
Acquisition	Scanner/vendor, field strength, sequence, views, voxel size, slice gap, temporal resolution, acceleration/reconstruction
Abnormality	Direction, magnitude, location, width/extent, transmurality and timing range
Estimator	Name/version, inputs, user interaction, representation and regularisation
Estimand	Reference, strain measure, axes, layer, dimensionality, region, temporal summary and aggregation
Reference	Simulation/phantom, DENSE, tagging, landmarks, tissue target or outcome; registration and uncertainty
Primary loss	Attribute-specific error at subject level
Tolerance	Clinically/technically justified threshold and sensitivity analysis
Splits	Participant-, centre- and time-aware separation; no final-test selection
Subgroups	Scanner, centre, image quality, pathology, wall thickness, abnormality scale and demographic groups
Failure policy	Exclusion, imputation, quality control or abstention; observable signals and coverage
Statistics	Bias/limits, confidence intervals, repeated-measures model, multiplicity and missingness

Supplementary Note 6: software, versioning and regulatory context

Feature-tracking and strain tools change over time. Regulatory clearance may cover a broad software application or specified outputs but does not establish material-motion accuracy for every regional analysis. Product webpages document current availability; they do not document the historical algorithm behind a published result. Studies should therefore record software name, exact version, analysis module, contour convention, manual corrections and any export transformation. Reference intervals and serial thresholds should not be transferred across products without a bridging study [5, 6, 18].

This review did not audit historical FDA summaries or manufacturer webpages. Its bibliographic citations and dated notes support only the limited interpretation above and do not constitute a current market survey. Authors should verify any product indication, clearance, version and jurisdiction against the live primary record before making a clinical recommendation.

Supplementary Note 7: reconstruction and dataset scope

Accelerated reconstruction can alter temporal sharpness, noise texture and edge appearance, all of which may affect downstream motion estimation even when conventional image-quality scores remain acceptable. Primary evidence is encouraging but endpoint-specific. In REGAIN, global feature-tracking strain from

49 participants showed near-zero mean differences between separately acquired accelerated and GRAPPA breath-hold cine, with non-zero limits of agreement [19]. CineVN's paired 18-volunteer comparison found the smallest strain bias among the tested accelerated reconstructions, but all accelerated protocols retained systematic peak radial/circumferential bias; the 46-patient demonstration had no conventional reference and included segmentation-related slice exclusions [20]. In ten healthy volunteers, model-based DLR reduced the dispersion of global circumferential strain differences from full sampling at high reduction factors relative to compressed sensing, while bias and radial underestimation remained [21]. DENT improved image-phase interpolation and reader-rated temporal presentation across a large multicentre development set and prospective testing, but it did not evaluate myocardial strain or validate its deformation vectors as tissue motion [22]. These measured findings motivate joint reconstruction–estimator tests; they neither show nor imply that deep learning erases focal abnormalities.

A reconstruction study that agrees on volumes, global strain or image appearance does not automatically validate regional correspondence, lesion location, spatial extent or timing. Ideally, the reconstruction and motion estimator are evaluated jointly against a material-sensitive or prescribed-motion reference, with the fully sampled or conventional reconstruction treated as a comparator that carries its own error. Temporal blurring, sharpening, alias suppression and hallucination are candidate mechanisms; none is an established explanation for a regional result until it is tested directly.

CMRxRecon2025 is useful for reconstruction robustness because the official site lists multiple centres, vendors, disease groups and a broad set of CMR modalities [23]. The official description does not provide material-motion ground truth, and it does not justify assuming that cine and LGE are paired for every participant. The 2024 challenge included facts and tasks that should not be transferred to 2025. Dataset labels, access, pairing, licensing and intended task must each be verified before use.

Supplementary Note 8: study-level interpretation examples

Study	Strongest direct support	Important positive evidence	What remains open
StrainNet [4]	DENSE-contour pixelwise displacement EPE; separately, cine-contour end-systolic global/segmental Ecc	Higher global and segmental Ecc agreement than conventional FT in 59 internal-test participants	EPE is not a cine-contour result; focal location/extent, other directions, external domains and reference uncertainty
Militaru et al. [5]	Product-specific global/regional agreement with tagging and LGE discrimination	Strong global LS/CS agreement for tested products	Regional/radial interchangeability and material truth
Kihlberg et al. [8]	Scar-discrimination performance in suspected CAD	DENSE/tagging outperformed FT for high-transmurality LGE segments	Pointwise motion error and other diseases/thresholds
Strain-8 [24]	Same-day precision by modality/direction/field strength	Several global measures had good or excellent repeatability	Accuracy, disease cohorts and longer-term reproducibility
MRXCAT2.0 [3]	Technical capacity/error under generated cases	Known motion paired with realistic-looking cine	Unmodelled anatomy, artefact, scanner and disease complexity

Jahromi et al. [25]	Phantom and healthy-volunteer component-specific comparison	Circumferential recovery was stronger than other directions in tested setting	Pathology, base/apex, 3D effects and clinical thresholds

Supplementary Table S1: empirical validation contracts

Table S1. Detailed measurement contracts for selected empirical studies. Unextracted or unverified items describe the limits of this review; they do not indicate absence from the original study.

Study and denominator	Population, input and analysis unit	Output, scale, layer, reference state and time	Reference, statistic and result	Directly supported inference
Wehner et al. [7]; 88 participants; 186 paired slices; torsion and dyssynchrony subset 38	Healthy and heterogeneous cardiac diagnoses from two centres; bSSFP plus matched DENSE; subject and slice	Transmural whole-slice base/mid/apex peak Green–Lagrange Ecc; four-chamber longitudinal strain; torsion and curve-delay dyssynchrony, ED reference	DENSE; biases in absolute percentage points: mid Ecc −0.4, CoV 11%; base −4.0, CoV 18%; apex −2.4, CoV 15%; dyssynchrony bias 26 ms, CoV 76%	Mid-slice peak Ecc agreement under matched definitions; poorer basal/apical magnitude and temporal synchrony agreement
Cao et al. [6]; 95 recruited, 87 analysed (77 cardiomyopathy, 10 controls)	Matched tagging, DENSE and bSSFP analysed by three FT products; subject and one matched mid slice/4-chamber	Peak Lagrangian Ecc and Err in one mid short-axis slice; Ell in four-chamber; systolic peak	HARP tagging; DENSE Ecc ICC 0.778, bias 0.155 percentage points, 95% limits −6.166 to 6.476; all methods had poor Err agreement	Relatively close Ecc but technique-dependent Err/Ell; no focal location, extent or dense-field endpoint
Militaru et al. [5]; 61 participants	18 healthy, 18 chronic MI, 15 DCM, 10 LV hypertrophy; bSSFP with three FT products, 3 T; subject and AHA segment	Global and 16-segment transmural peak-systolic Lagrangian LS, CS and RS; ED reference	HARP tagging; global ICC 0.92–0.94 LS, 0.88–0.91 CS; intervendor ICC 0.10–0.81 for global RS (FT–FT, not FT–tagging); LGE discrimination tested in MI	Strong global LS/CS agreement for tested products; regional/radial values and scar discrimination are product dependent
Kihlberg et al. [8]; 116 patients	Suspected CAD; DENSE, tagging, bSSFP FT and LGE; subject and segment	ROC endpoint: end-systolic segmental circumferential strain; other global/segmental strain and torsion analyses were also reported	LGE > 50% transmurality; segment AUC 0.87 DENSE, 0.83 tagging, 0.66 FT; sensitivity at 80% specificity 82%, 71%, 35%	Technique-specific discrimination of high-transmurality scar; not a dense displacement ranking
StrainNet [4]; 305 total, 243 development/670 sections, 62 internal test/190 sections; cine analysis 59	Patients, healthy adults and healthy children/adolescents; DENSE magnitude contours for displacement test, separately bSSFP cine FT contours; pixel, subject and AHA segment	Intramyocardial displacement over frames; separately end-systolic global/segmental circumferential Green–Lagrange strain, ED reference	DENSE phase-derived displacement; EPE 0.75 ± 0.35 mm with DENSE contours; cine Ecc ICC 0.87 global/0.75 segmental versus FT 0.72/0.48; segmental LoA −9.16 to 8.07 percentage points	Positive internal-test displacement performance under DENSE-contour input and bounded cine end-systolic Ecc agreement; the two tests are not interchangeable
Strain-8 [24]; 20 healthy volunteers	Same-day rescan using cine, tagging and DENSE at 1.5/3 T plus 2D/3D echo; subject and segment	Global and AHA-segmental frame-wise peak strain; modality-specific layer/definition	Repeat scan; CoV, ICC and Bland–Altman; global CoV ≤ 20% across protocols, segmental values frequently > 20%, especially longitudinal/radial	Precision and minimum detectable change under controlled same-day conditions; no accuracy or disease-cohort claim

Study and denominator	Population, input and analysis unit	Output, scale, layer, reference state and time	Reference, statistic and result	Directly supported inference
Jahromi et al. [25]; deforming phantom and 40 healthy volunteers	One-slice cine model versus one-/two-slice DENSE models; subject, slice and endocardial/epicardial layer	Slice-wise Ecc, Ell and Err through the normalised cycle; midventricular; model-specific reference	Full-text results: peak median systolic endocardial Ecc, cine minus two-slice DENSE −2.7 percentage points; TOST within selected ±3.5-point margin; slice-clustering treatment unclear	Endocardial Ecc equivalence in healthy mid-slice setting; not base/apex, disease or focal lesion validation

For StrainNet, the Results prose reports global cine–DENSE limits of agreement of −5.74 to 4.72 percentage points, whereas Table 2 gives bias −0.51 with a 95% limits half-width of 5.15 percentage points. These do not reproduce exactly the same interval. The global interval in the main text follows the source prose; the segmental interval is consistent across the two presentations. This small source discrepancy is retained as reported; no silent reconciliation was attempted [4].

Supplementary Table S2: validation-resource details

Table S2. Validation resources and reference conditions. Descriptions combine the October search records and targeted checks on 11 October 2026; unresolved access is retained. The phrase not file-verified means that no subject-level archive was downloaded or processed in this review.

Resource	Nature, scale and population	Modalities, coverage and pairing	Motion reference and evaluable scale	Access, processing requirement, uncertainty and domain boundary
STRAUS [26, 27]	Synthetic: 18 cases generated from 3 volunteer templates, each with healthy, LBBB, LCX, LAD, RCA and RCA2 states	For each case: 3D cine MR, three tagged-MR sequences and 3DUS over the cardiac cycle; 90 sequences/2700 volumes reported	Framewise reference meshes provide dense correspondence; the paper reports regional engineering strain	Public/no login reported; MHD/RAW images and VTK meshes are ready files but require matched coordinate/time/reference handling; imposed mechanics, three anatomies and simulated appearance bound transfer
Stanford Cardiac Performance MRI [28]	Human: laboratory page 51 healthy; DOI metadata 55 total (51 healthy, 4 aortic stenosis), acquired 2023–2025	3 T whole-LV bSSFP cine; DENSE at corresponding locations with xyz encoding; three mid-LV tagged slices; T1 and cDTI described	DENSE phase and tagging can support in-vivo displacement/strain comparisons after processing; whole-LV cine coverage does not imply whole-LV tagged reference	Stanford Digital Repository, ODbL 1.0; subject ZIP DICOMs; not file-verified here; requires reconstruction/phase unwrapping, segmentation, registration and estimand alignment; cohort total is not a usable-pair count; Jahromi reports using this resource, but subject-file linkage is not verified
CMAC [29, 30]	Dynamic phantom and 15 healthy volunteers; challenge database	Cine SSFP, 3D tagged MRI and 3DUS; whole-cycle 3DTAG and cine with modality registration described	12 landmarks (four walls at three LV levels) tracked by two observers in 3DTAG; sparse point error, not dense field truth	Open access with attribution; median inter-observer variability 0.77 mm phantom and 0.84 mm volunteers; healthy/sparse/legacy acquisition domain
MRXCAT [31]	Synthetic generator based on XCAT; not a fixed participant cohort	Cine and first-pass perfusion with cardiac/respiratory motion, multiple coils, noise, trajectories and undersampling	Prescribed XCAT motion and fully sampled simulation support sampling/reconstruction and known-motion tests	MATLAB framework available for non-commercial research by request; XCAT access and self-generation required; signal, texture, partial-volume and anatomy assumptions limit realism

Resource	Nature, scale and population	Modalities, coverage and pairing	Motion reference and evaluable scale	Access, processing requirement, uncertainty and domain boundary
MRXCAT 2.0 [3, 32]	Synthetic generator, not a fixed cohort; healthy, infarct, DCM and HCM exemplars	bSSFP-like cine generated from statistical LV shape, biophysical function and texturised tissue labels	Known simulated function/strain enables voxel/mesh-scale capacity and estimator tests	Repository cited by the paper; live code/example access was blocked during this check; generation and dependencies require verification; generated cases are not independent patients and model/texture coverage bounds transfer
CMRxRecon 2025 [23]	Human multicentre, multivendor, multidisease reconstruction challenge/resource; exact motion-reference denominator not applicable	Cine plus mapping, contrast-weighted, black-blood, phase-contrast, LGE and perfusion described; pairing varies by task/release	No declared dense material-motion truth; can assess reconstruction robustness and downstream domain shift if endpoints are added	Challenge terms and released reconstruction/QC files; disease labels are not motion truth and cine–LGE pairing cannot be assumed for every participant

Supplementary Table S3: cine-CMR motion estimators by year

Table S3. Motion and strain estimators discussed in the main text, in approximate chronological order within each family. Summaries retain the source-specific input, representation and available validation description. NV means not verified at the available reading depth; it is not an absence claim. Most lineage entries use abstracts; full-text corrections to central studies are identified in the main text and evidence ledger. Rows are not ranked and differ in status (journal, conference, workshop, preprint); the year/source column states which is which. A short validation entry usually means that the method was evaluated for the claim its own authors made; it does not mean that the method was tested and found wanting for this review's question.

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Conventional estimators and comparators					
Boundary feature tracking	Family [33]	Cine plus endo/epicardial contours; per-case tracking	Contour continuity and propagation; interior motion largely interpolated	Many clinical cohorts; inter-software/version agreement, scan–rescan, tagging/DENSE comparison, clinical association	Many disease cohorts; regional and radial agreement consistently weaker than global [5, 24, 34]
Free-form deformation (FFD)	1999; IEEE TMI [35]	Image pair; none	Affine plus B-spline lattice with smoothness cost; NMI data term	Contrast-enhanced breast MRI; comparison with rigid/affine registration	Not applicable (non-cardiac origin); transformation model reused by later cardiac work
Harmonic phase (HARP)	2000; IEEE TMI [36]	SPAMM-tagged MR (comparator, not cine); none	Phase of an isolated spectral peak	Real and simulated tagged images; small-motion displacement and 2D strain	Not applicable; material-sensitive comparator
Early clinical feature tracking	2009–2013 [37–39]	Cine plus contours; vendor feature tracking, per case	Proprietary feature tracking	30 post-STEMI patients (scar association); 191 DMD patients and 42 controls versus HARP, mid-LV whole-slice Ecc; 145 healthy volunteers with tagging in 20 (abstract level)	Scar association and whole-slice/global agreement; focal localisation endpoint NV
Groupwise nD+t B-splines	2011; Med Image Anal [40]	Whole dynamic sequence; none	Spatio-temporal B-spline, optional cyclic constraint, no fixed reference	Synthetic and 3D+t lung/cardiac CT; sub-voxel accuracy; cardiac CTA volume curves with mean deviation about 3% from manual curves (abstract)	NV

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Laplace-based contour mapping	2015; Int J CARS [41]	Cine contours; none	Laplace equation between successive contours	Synthetic data and in-vivo feasibility (abstract level)	NV
Variational data assimilation	2016; IEEE TMI [42]	Cine sequence; none	Contraction model coupled to image observations	Known-motion synthetic data; 47 patients at three slice levels; tagging strain ICC above 0.8 and clinical-score associations (abstract)	NV as a prescribed focal experiment
Gradient-based optical flow	2019; Med Image Anal [43]	Cine and tagged MR; none	Intensity and shape-feature invariance, iteratively optimised	Realistic simulations and preliminary clinical data (abstract level)	NV
Automated segmentation plus spline tracking	2017; FIMH [44]	Cine; supervised segmentation, unsupervised tracking	Learned segmentation feeding a conventional tracker	42 HCM patients and 21 controls; pixel accuracy about 97% against manual segmentation	Group separation HCM versus controls; material-motion reference and focal experiment NV
Automated segmentation plus feature tracking	2018; ISBI [45]	Short- and long-axis cine; supervised segmentation	CNN segmentation followed by conventional feature tracking	40 healthy volunteers and 40 ischaemic patients; strain agreement with cvi42 (abstract)	Inter-method agreement; material-motion reference and focal experiment NV
Anatomy-Aware Tracker	2020; MLMI [46]	Long-axis cine; weak shape supervision	Convolutional VAE of realistic shapes refining a dense tracker	Tracking versus other methods; qualitative shape realism	NV
Groupwise locally low-rank	2024; BMC Med Imaging [47]	Whole cine sequence; none	Groupwise registration minimising patchwise signal rank	Simulation ($n = 20$), public cine ($n = 100$), in-house tagging patients ($n = 16$); point, voxelwise and global strain error; late-diastolic drift; tagging radial-strain correlation	Paired tagging in patients; disease composition NV; prescribed focal test NV
Unsupervised learning-based registration					
VoxelMorph	2019; IEEE TMI [48]	Image pair; unsupervised (optional auxiliary segmentations)	Amortised similarity-plus-smoothness objective	General registration datasets; accuracy and run time versus selected comparators	NV
TransMorph	2022; Med Image Anal [49]	Volumetric pair; unsupervised	Hybrid Transformer–CNN; diffeomorphic and Bayesian variants	Brain MRI (inter-patient, atlas) and phantom-to-CT; topology preservation; uncertainty calibration	NV for cardiac pathology
Fourier-Net	2023; AAAI [50]	Image pair; unsupervised	Band-limited base representation; inverse Fourier decoder; diffeomorphic integration changes final spectral support	Registration benchmarks; accuracy with speed, memory and multiply-adds	NV
Fourier-Net+	2026; IEEE TNNLS [51]	Band-limited image representation; unsupervised	As Fourier-Net with shortened contracting path; cascaded variant	Five datasets including one 3D cardiac MRI set; Dice, Hausdorff distance, ED volume and EF versus TransMorph and other baselines at lower memory and compute (abstract level)	NV; cardiac evaluation is geometric/functional, not regional strain
Cardiac-specific estimators					
Joint motion and segmentation	2018; MICCAI [52]	Cine sequence; unsupervised motion, weakly supervised segmentation	Shared multi-scale encoder; Siamese recurrent spatial transformer	220 total subjects, including 100 test subjects; mask propagation and speed versus competing methods	NV

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
Apparent flow	2019; Med Image Anal [53]	Two cine phases; semi-supervised	Pixel-wise flow combined with masks	ACDC; nine shape/volume/EF and motion features; 95% five-fold cross-validation and 94% held-out challenge test accuracy	Pathology classes are the target; motion validated via the downstream label
Motion Pyramid Networks	2020; MICCAI [54]	Cine pair; unsupervised, cyclic teacher–student	Multi-scale fused pyramid	Two public datasets; mask metrics and errors against known motion applied by a separate network; inference time	Model-derived imposed motion; not independent physiological truth
FOAL	2020; CVPR [55]	Cine sequence at test time; meta-learned online optimiser	Test-time adaptation of a learned tracker	Two public datasets; accuracy under train–test mismatch; about 0.4 s per video for online optimisation	Distribution shift is the target, not characterised as focal abnormality
MulViMotion	2022; IEEE TMI [56]	SAX plus 2-/4-chamber LAX cine; unsupervised with weak shape supervision	Hybrid 2D/3D network; shape regularisation module	580 UK Biobank subjects: 450 training, 50 validation, 80 test; 3D LV motion comparison	NV
DeepMesh	2024; IEEE TMI [57]	SAX and LAX cine; unsupervised via differentiable rasteriser	Template 3D endo/epicardial mesh propagated to subject space	UK Biobank; vertex-wise 3D displacement with temporal correspondence and common template indexing	NV
Bidirectional registration network	2023; Comput Biol Med [58]	Cine sequence; unsupervised	Bidirectional reference-to-frame prediction with sequence context	Public cine data (abstract level)	NV
MCM (Mamba)	2025; MICCAI workshop [59]	Cardiac-cycle image sequence; unsupervised	Bidirectional Mamba block; adjacent-frame motion decoder	Two public datasets; comparison with conventional and learned trackers	NV
SegMorph	2025; IEEE TMI [60]	Cine sequence; joint motion and segmentation	Recurrent variational representation	Public cine data (abstract level)	NV
CardioMorphNet	2026; Med Image Anal [61]	SAX cine volumes; recursive registration of segmentation maps without an intensity loss	Recurrent VAE with separate segmentation and motion posteriors; uncertainty map	UK Biobank and M&Ms; warped versus reference masks; uncertainty versus other probabilistic methods	M&Ms is multi-centre/multi-pathology, but the comparison is mask agreement, not focal motion accuracy
Bidirectional latent neural ODE	2026; Comput Med Imaging Graph [62]	Cine sequence; unsupervised	Continuous-time latent ODE; Lagrangian consistency	ACDC and M&Ms; temporal consistency and plausibility versus selected comparators	NV
STCMT-Net	2026; Med Image Anal [63]	4D CT and cine MRI; unsupervised	Unsupervised keypoints as anchors; dense field from matched-keypoint basis vectors	4D CT and cine datasets; comparison with competing methods; strain analysis separating cardiac conditions	Group separation by condition; prescribed focal test NV, material-motion reference NV
Semi-supervised motion flow	2025; Comput Biol Med [64]	Full cine sequence; base, distance-map and fully supervised configurations	Memory network over all frames	Abstract: 271 patients; EPE 1.02 mm base, 0.95 mm distance-map, 0.91 mm full supervision; RAFT 1.19 mm; global peak RS/CS correlation 0.83/0.90	EPE reference: manually initialised contours propagated by conventional FT; not dense material truth; focal endpoints NV
IG-BiReg	2026; Phys Med Biol (online first) [65]	Input/supervision details NV	Title-level implicit-guided bidirectional residual registration	Primary scientific content not accessed in finalisation; performance NV	NV

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
CMR foundation models	2025–2026 [66, 67]	36M images/27,524 subjects (self-distillation); 15M cine images/74,916 studies	Pre-trained representation fine-tuned per task	Segmentation, landmark localisation, diagnosis, prognostication across independent datasets	Dense motion was not among the tasks in the retrieved summaries; other task validation was reported
Physics-, biomechanics- and representation-constrained estimators					
Biomechanics-informed prior	2023; Med Image Anal [68]	Cine sequence; unsupervised, prior learned from simulated deformations	VAE manifold of plausible deformations traversed at tracking time	Three public cine datasets; tracking accuracy, volume preservation, generalisability, derived strain	NV as a prescribed focal experiment
WarpPINN	2023; Med Image Anal [69]	Cine bSSFP; per-case optimisation	Neural deformation with hyperelastic regulariser and Jacobian near-incompressibility; Fourier features	Synthetic examples; 15-volunteer cine benchmark; landmark tracking; strain by automatic differentiation	Healthy benchmark; focal abnormality not evaluated
WarpPINN-fibers	2026; Eng Comput [70]	Cine MR; per-case optimisation	As WarpPINN plus fibre-stretch penalty along synthetic fibres	Known-motion synthetic phantom; 15 healthy participants: landmark comparison and strain-curve plausibility without reference strain curves	Healthy benchmark; focal abnormality not evaluated
Space–time finite-element correlation	2026; Med Image Anal [71]	Routine SAX and LAX cine; correlation framework	Region-specific biomechanical regularisation; data-driven temporal decomposition; multi-view alignment	2D+t: synthetic known motion/strain, three public datasets and one clinical dataset; 3D+t: clinical masks only; two FT and four DL comparators (abstract)	One synthetic ground-truth setting; in-vivo references are landmarks and masks; size/severity sweep NV
Reference-supervised and end-to-end strain					
Cine-only infarct delineation	2018; Med Image Anal [72]	Non-contrast cine; supervised on manual LGE delineation	LSTM motion features plus optical flow, fused per pixel	165 cine datasets; pixel accuracy 95.0%, Dice 89.9%, $\kappa = 0.91$ against two radiologists	Infarction is the target; reference is LGE tissue, not motion
Survival-supervised motion	2019; Nat Mach Intell [73]	RV cine-derived motion; supervised on survival in chronic thromboembolic pulmonary hypertension	Learned low-dimensional motion representation	302 patients; internally bootstrap-corrected Harrell C 0.75 (95% CI 0.70–0.79) versus conventional manual-variable model 0.59 (0.53–0.65)	Prognostic discrimination in chronic thromboembolic PH; not an LV focal-motion or management-impact endpoint
DeepTag	2021; CVPR [74]	Tagged MR (comparator acquisition); unsupervised	Bidirectional diffeomorphic registration; differentiable composition to Lagrangian field	Clinical tagging dataset; landmark tracking accuracy and efficiency	NV
3D tagging networks (synthetic training)	2025; JCMR [75]	3D tagged MR; synthetic training from a biophysical LV model	Two networks predicting LV motion through the cycle	Synthetic validation plus external human manual-landmark comparison and porcine strain comparison with manual cine annotations	Tagging analysis, not routine cine; manual reference has its own uncertainty
DeepStrain	2021; Front Cardiovasc Med [1]; 2023 comparison (online 2022) [2]	Cine bSSFP; image, anatomical-label and smoothness losses	Segmentation plus motion network; end-diastolic reference; Green–Lagrange strain	ACDC ($n = 150$); CMAC landmark EPE 2.9 ± 1.5 mm; scan–rescan in ten volunteers; 47 healthy/533 patients versus manual FT, r 0.83–0.85 radial and 0.88–0.91 circumferential	Independent MRXCAT2.0 evaluation on normal, dilated, hypertrophic and infarcted cases (main text, known-motion tests)

Method	Year; source	Input; supervision	Prior or representation	Reported data and validation	Evaluation on pathology or focal abnormality
StrainNet	2023; Radiol Cardiothorac Imaging [4]	Contour time series (DENSE-magnitude at displacement test; cine FT at strain test); supervised on DENSE displacement	2D+t CNN; prior implicit in DENSE-derived training distribution	243 development/62 internal-test participants (670/190 sections); EPE 0.75±0.35 mm with DENSE-magnitude contours; cine Ecc ICC 0.87 global/0.75 segmental versus FT 0.72/0.48; segmental LoA −9.16 to 8.07 percentage points in 59	Mixed patient population with material-sensitive reference; size/severity sweep NV; no focal location/extent endpoint
DL synthetic strain	2023; Radiol Cardiothorac Imaging [76]	Cine at inference; training targets: coregistered 4D-flow velocity and commercial-software segmentation	Velocity and segmentation network; strain derived from estimated motion	Development: 223 exams/219 patients; clinical test: 53 patients/846 segments, four-reader WMA consensus; ROC AUC 0.90; 30% peak Err threshold: sensitivity 86%, specificity 85%	WMA clinical endpoint; reader labels evaluate the output, not the motion-training target
DENSE-guided rotation network	2025; IEEE ISBI [77]	Routine bSSFP cine; supervised on DENSE displacement	Separate rotation and radial branches fused	NV in the record retrieved; strain accuracy against DENSE with large rotation as target	NV
DL aligned strain	2025; Radiol Artif Intell [78]	Cine key phases and segmentation; deformation loss: SSIM, smoothness, Dice	Deformation-derived strain aligned for downstream classification; LGE labels classify patients	139 DMD patients: 57 development, 82 same-centre temporal holdout; endpoint: at least one LGE-positive segment	Patient-level fibrosis classification, not scar-boundary error or LGE-supervised motion
MyoNet	2026; Bioengineering [79]	Binary spatiotemporal inputs from cine segmentation; displacement targets from SinMod on paired tagging	CNN alternating spatial and temporal convolution	Dahl salt-sensitive rats after radiation; ICC 0.973 circumferential, 0.975 radial	Preclinical tagging-reference agreement; not human cine validation

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